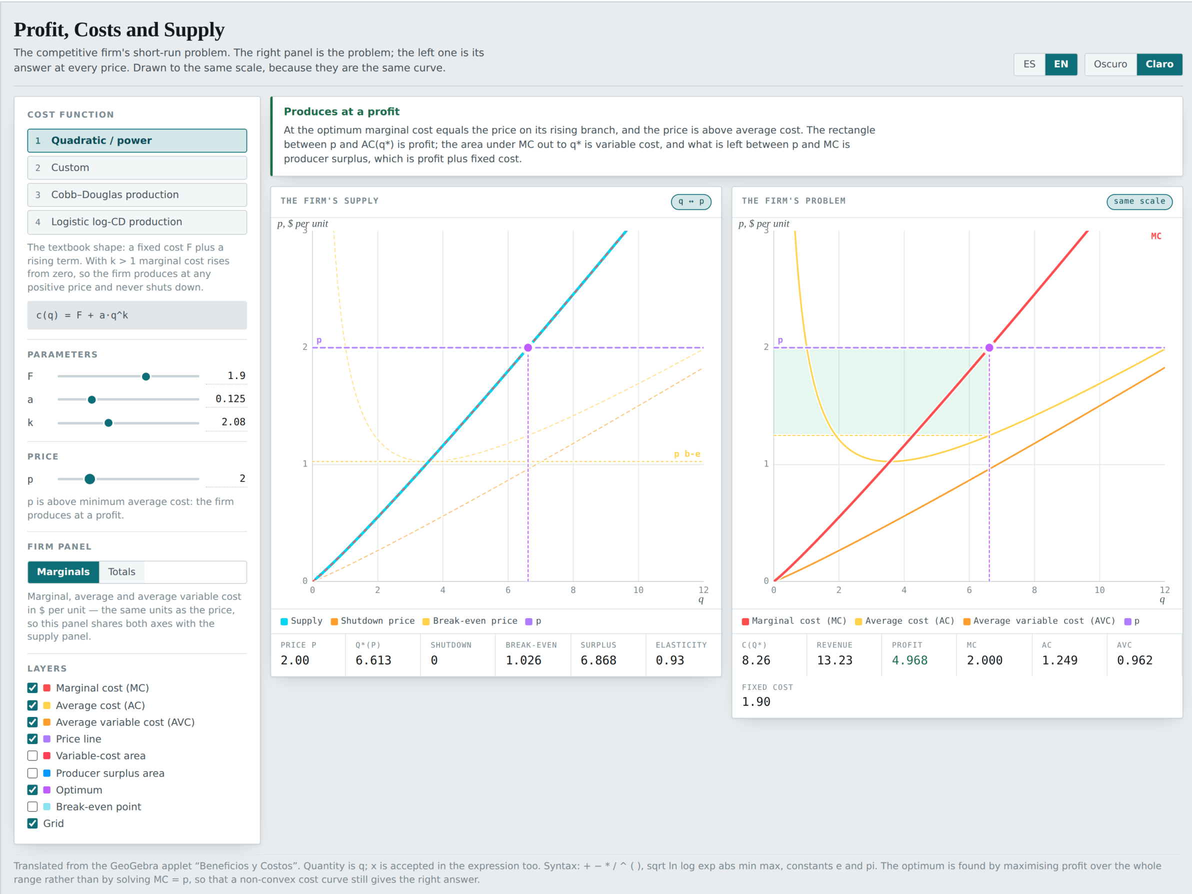
MANUAL

Profit, Costs and Supply

The competitive firm’s short-run problem, and the supply curve that comes out of it. Two panels drawn to the same scale, because they are the same curve.



Applet manual · Introductory Microeconomics · Universidad de los Andes
Translated from the GeoGebra applet “Beneficios y Costos”.
 It opens in a browser; there is nothing to install.

Contents

1 What this is

A competitive firm in the short run has exactly one degree of freedom: how much to produce. Capital is fixed, the price is handed to it, and all it can do is pick the quantity q that maximises profit

$$\pi(q) = pq - c(q).$$

That is the problem. The firm's supply curve is the *answer* to that problem, solved once for every possible price. They are two different objects, and in almost every textbook they appear in two different figures on two different pages, joined by a sentence saying that marginal cost "is" the supply curve, which the student has to take on trust.

This applet puts them side by side and to the same scale. The right panel is the problem; the left one is its answer. Marginal cost is drawn in both, in the same coordinates, and in the supply panel it is laid over the supply curve as a red dashed line. Above the shutdown price the dashes fall exactly on top. Nothing has to be taken on trust.

Figure 1. The two panels at the opening settings. Left: the firm's supply curve in the (q, p) plane. Right: the problem, with marginal, average and average variable cost. Both axes are the same in both panels, so marginal cost occupies the same place in each.

The point

Marginal cost and inverse supply are not similar curves: above the bottom of average variable cost they are the same curve, point for point. The whole applet is built so that this can be seen rather than asserted.

1.1 Who it is for

A first course in microeconomics, at the session where profit maximisation turns into the supply curve. It assumes the student has already met total, average and marginal cost, and knows what an optimum is. It assumes no calculus beyond the idea of a slope.

2 Opening it

There is nothing to install. The whole application is one `index.html` file: it downloads no libraries, calls no server, and stores nothing outside your browser.

2.1 The three ways

Double-click the file. The quickest route. Find `firm-supply/index.html` and open it. The browser shows it straight off the disk, with an address beginning `file://`. It works offline from the first moment.

From the web. If someone has published it, the address ends in `/firm-supply/`. Once the page has loaded you can disconnect: it needs the network for nothing else.

Served from a folder. For a class, putting the folder on any static server is enough. With Python installed:

```
python3 -m http.server 8000
```

then `http://localhost:8000/firm-supply/` in the browser.

> **Note**
>
> Saving the page with `Ctrl` + `S` (or `Cmd` + `S` on a Mac) leaves a copy of your own that still works offline and that can travel on a memory stick or go out by email.

2.2 Language and theme

Two pairs of buttons sit at the top right.

Control	What it does
ES / EN	Switches the whole interface between Spanish and English on the spot. Nothing is recomputed, so you can switch mid-explanation without losing the state you had.
Dark / Light	Switches the theme. Dark is the original design; light exists for projectors and for printing.

Both choices are remembered in the browser, so next time it opens the way you left it.

2.3 Which browser

Any reasonably recent one: Chrome, Edge, Firefox or Safari, on a computer, tablet or phone. On a narrow screen the panels stack one above the other instead of sitting side by side. No account, no permissions, no extensions.

3 The screen

Left to right, top to bottom:

The control rail. Picks the cost function, moves its parameters, sets the price, decides what the firm panel shows and which layers are drawn. Every slider has a numeric box beside it: drag, or type the exact number.

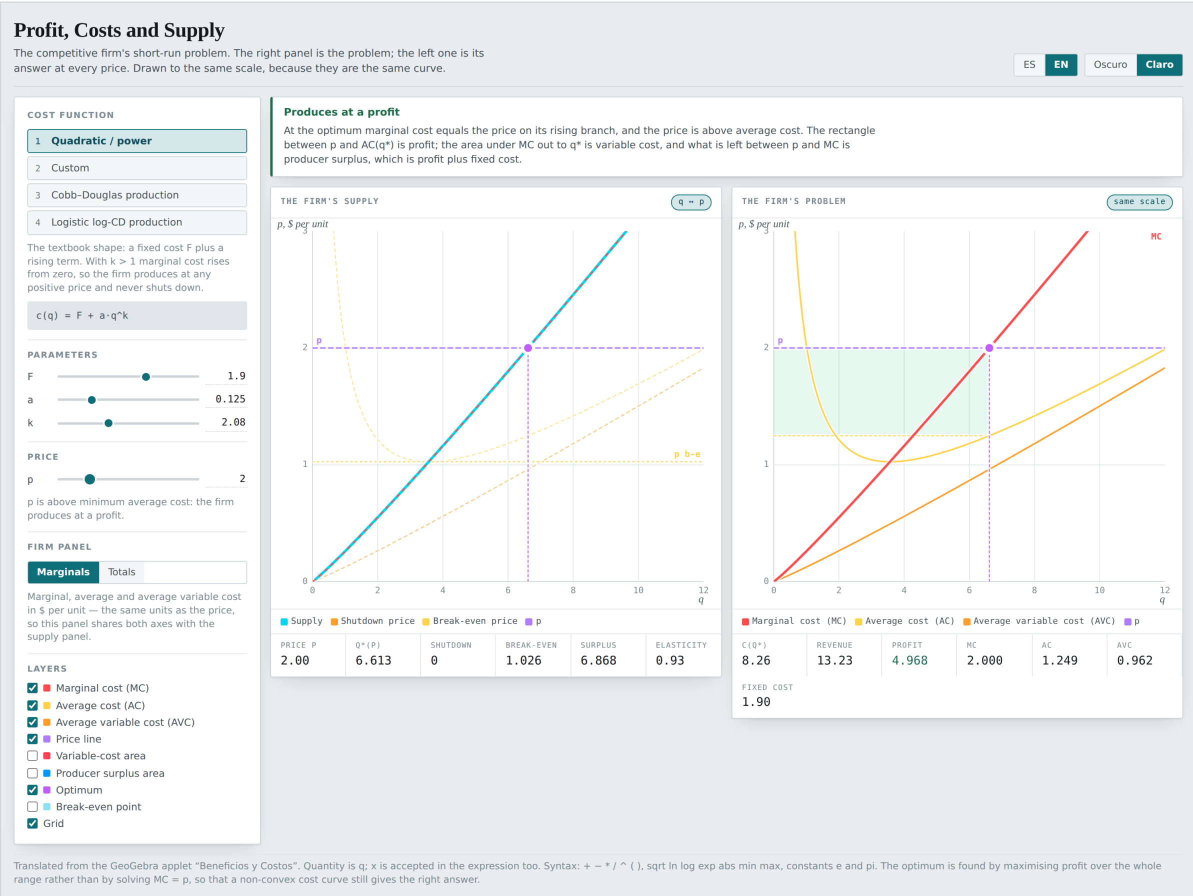


Figure 2. The whole screen. Controls down the left; the verdict along the top; the two panels with their readouts below.

The verdict. The coloured band under the title. It names which case the firm is in — profit, loss, shutdown, break-even, capacity, no optimum — and explains why. It changes on its own as you move anything.

The supply panel (left). The inverse supply curve in the (q, p) plane, with the shutdown price, the break-even price and the current price.

The firm panel (right). Either the marginal curves or the totals, depending on the **Firm panel** switch.

The readouts. The row of numbers under each panel. They are the same values that are drawn, written to three decimals for when the eye is not enough.

3.1 Dragging to change the price

In any panel whose vertical axis *is* the price — the supply panel always, the firm panel when it is showing marginals — you can drag with the mouse or a finger and the price follows the pointer. It is the natural way to read a supply curve: pick a price, see what quantity comes out.

In the totals view the vertical axis is money, not money per unit, and the panel does not answer the pointer.

3.2 Keyboard

Key	What it does
`1` `2` `3` `4`	Pick the cost function.
`m`	Firm panel to marginals.
`t`	Firm panel to totals.

3.3 The dark theme

Dark is the original design; light is there for projectors and for printing. The two were designed separately — it is not an inversion of the colours.

4 The model

4.1 Costs

Everything starts from a total cost function $c(q)$: what it costs to produce q units when the capital is already paid for. The rest follow from it.

$$
\begin{array}{rc}
\text{Fixed cost} & F = c(0) \\
\text{Variable cost} & VC(q) = c(q) - F \\
\text{Marginal cost} & MC(q) = c'(q) \\
\text{Average cost} & AC(q) = \dfrac{c(q)}{q} \\
\text{Average variable cost} & AVC(q) = \dfrac{c(q) - F}{q}
\end{array}
$$

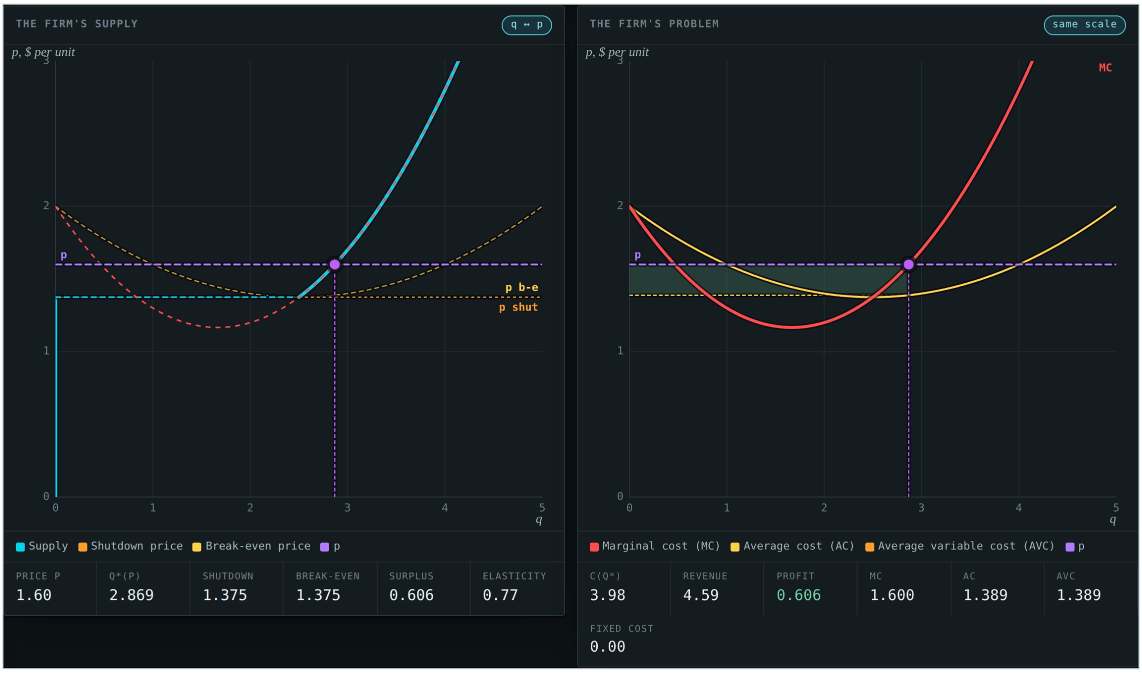


Figure 3. The same two panels in the dark theme.

Fixed cost is read as the limit of $c(q)$ as q falls to zero from above. In the families that start from a production function that limit is $r\bar{K}$, the rental on the fixed capital: it is paid whatever is produced.

4.2 The problem

$$\max_{q \geq 0} \quad \pi(q) \; = \; p\,q - c(q).$$

If the maximum is interior, the first-order condition gives $MC(q^*) = p$, and the second-order condition requires marginal cost to be *rising* there. Both matter: on a cubic cost curve the equation $MC(q) = p$ has two roots and one of them is a *minimum* of profit.

But $q^* = 0$ is a candidate too, and not a limiting case of the other one: it is a corner, with profit $\pi(0) = -F$. The firm produces only if producing beats closing.

4.3 The shutdown price

Comparing the two:

$$p\,q - c(q) \; > \; -F \quad \Longleftrightarrow \quad p\,q \; > \; c(q) - F = VC(q) \quad \Longleftrightarrow \quad p \; > \; AVC(q).$$

Fixed cost drops out of the comparison. It is already sunk, and a decision that takes it into account is a decision taken wrongly. Out of this comes the **shutdown price**:

$$\underline{p} \; = \; \min_{q>0} AVC(q),$$

and the rule everybody recites: the firm produces as long as the price covers average variable cost, even when it does not cover average total cost.

4.4 The break-even price

$$p_{be} \;=\; \min_{q>0} AC(q).$$

The price at which profit is exactly zero. Above it there is profit; below it there are losses — which may still be worth taking, as long as the price stays above $\underline{p}$.

4.5 Supply

$$q^*(p) \;=\; \begin{cases} 0 & \text{if } p < \underline{p}, \\ \text{the root of } MC(q) = p \text{ on the rising branch} & \text{if } p > \underline{p}. \end{cases}$$

At $p = \underline{p}$ the firm is indifferent, and supply **jumps**: it goes from zero to the quantity $\underline{q}$ that minimises average variable cost, without passing through the values in between. That jump is not a technicality. It is why the supply curve of a firm with fixed costs does not reach the origin.

4.6 Producer surplus

$$PS \;=\; \int_0^{q^*} \big(p - MC(q)\big)\, dq \;=\; p\,q^* - VC(q^*) \;=\; \pi + F.$$

One number written three ways, and the applet draws two of them: in the right panel as the area between p and marginal cost, and in the left as the area between the price axis and the supply curve, from $\underline{p}$ up to p. The two regions have different shapes — with a jump they cannot coincide — and exactly the same area.

> **Note**
>
> Producer surplus and profit are the same thing only when there is no fixed cost. With $F > 0$ the surplus is always larger, by exactly F.

5 The four cost functions

Picked with the four buttons at the top of the rail, or with `1` to `4`. Each brings its own sliders.

5.1 1 · Quadratic / power

$$c(q) = F + a\,q^k, \qquad k > 1.$$

The textbook shape. Marginal cost, $MC = akq^{k-1}$, rises from zero, so average variable cost has no interior minimum: its infimum is zero and is approached only in the limit. The consequence: **this firm never shuts down**. At any positive price it is worth producing something.

Slider	What it is
F	Fixed cost. Raises or lowers average cost without touching marginal cost.
a	Scale of variable cost.
k	Curvature. With k near 1 marginal cost is nearly constant and the problem stops having a solution; with k large it rises very fast and supply goes vertical.

5.2 2 · Custom

Type $c(q)$ into the text box. Marginal cost is obtained by differentiating the expression *symbolically*, not numerically, so it is exact. The one supplied by default,

$$c(q) = 0.1\,q^3 - 0.5\,q^2 + 2q,$$

is the textbook cubic: no fixed cost, U-shaped average variable cost with minimum 1.375 at $q = 2.5$, and therefore a strictly positive shutdown price and a visible jump in supply.

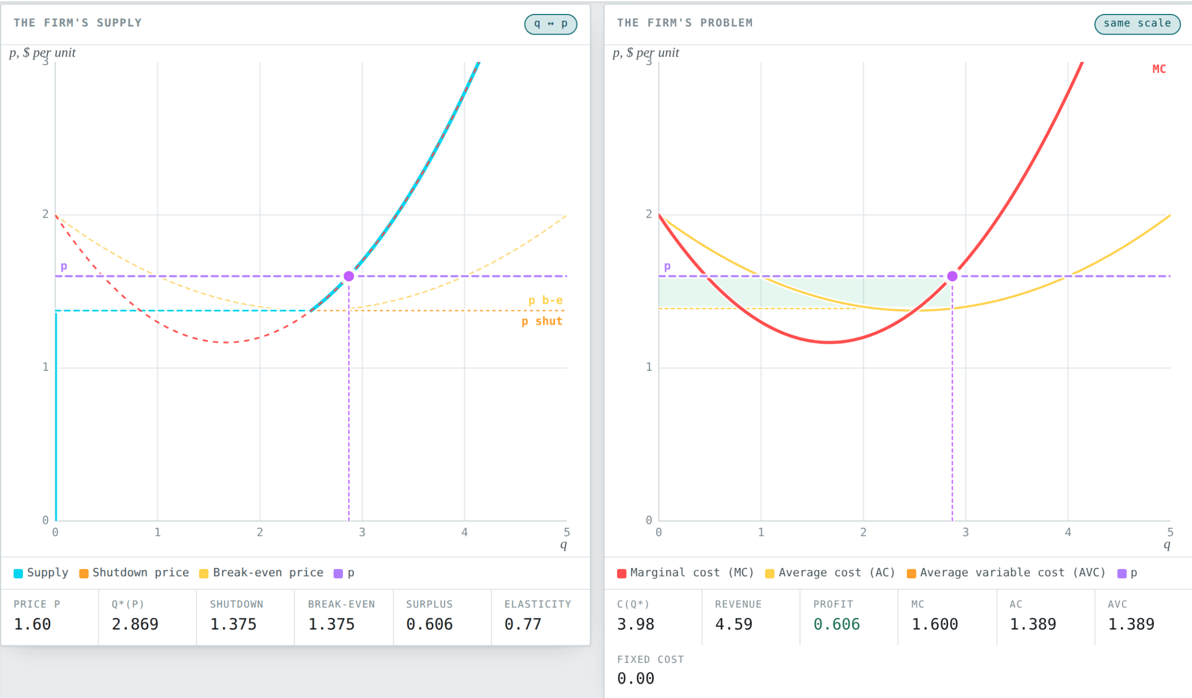


Figure 4. The cubic at $p = 1.6$. The supply panel shows all three pieces of the curve: flat against the price axis below 1.375, the discontinuous jump, and the rising branch that coincides with marginal cost.

5.3 3 · Cobb–Douglas production

In the short run capital is fixed at $\bar{K}$ and the only variable factor is labour. Starting from

$$f(K,l) = b\left(K^{\beta} l^{1-\beta}\right)^{s},$$

invert it for the labour needed to produce q,

$$l(q) = \left(\frac{q}{b\,\bar{K}^{\beta s}}\right)^{e}, \qquad e = \frac{1}{(1-\beta)s},$$

and price it at the wage w:

$$c(q) \;=\; r\bar{K} \;+\; w\, b^{-e}\, \bar{K}^{-\beta/(1-\beta)}\, q^{e}.$$

Everything turns on e. With $e > 1$ there are diminishing returns to the variable factor, marginal cost rises, and the problem has a solution. With $e \leq 1$ marginal cost does not rise, every extra unit adds profit, and there is no optimal quantity. The applet says so instead of drawing something meaningless.

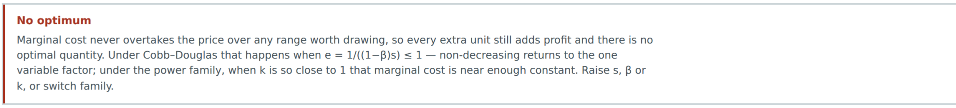


Figure 5. The verdict when $e \leq 1$: no optimum, and what to move so that there is one.

Slider	What it is
b	Total factor productivity.
β	Capital's elasticity. It enters e.
s	Returns to scale of the Cobb–Douglas. It enters e.
w	The wage. Scales all of variable cost.
r	The rental on capital. It is the fixed cost, $F = r\bar{K}$.
$\bar{K}$	The fixed capital. It is what makes the short run short.

5.4 4 · Logistic log-CD production

The same idea, with a production function that *saturates*:

$$g(K, l) = \frac{b\,K^{\gamma}}{1 + e^{-s(\ln f - m)}},$$

whose ceiling is $\bar{Q} = b\,\bar{K}^{\gamma}$ however much labour is hired. Inverting it,

$$c(q) \;=\; r\bar{K} \;+\; w\, e^{m/(1-\beta)}\, \bar{K}^{-\beta/(1-\beta)} \left(\frac{q}{\bar{Q} - q}\right)^{e}, \qquad 0 < q < \bar{Q}.$$

Marginal cost blows up as $\bar{Q}$ is approached, and the supply curve has a vertical asymptote there. It is the honest picture of what a fixed capital means: however far the price rises, there is a quantity this plant cannot pass.

6 The controls

6.1 Price

A slider from 0 to 10, with a numeric box for an exact value. Under it a sentence says where the price stands relative to the shutdown and break-even prices. It can also be moved by dragging inside any panel whose vertical axis is the price.

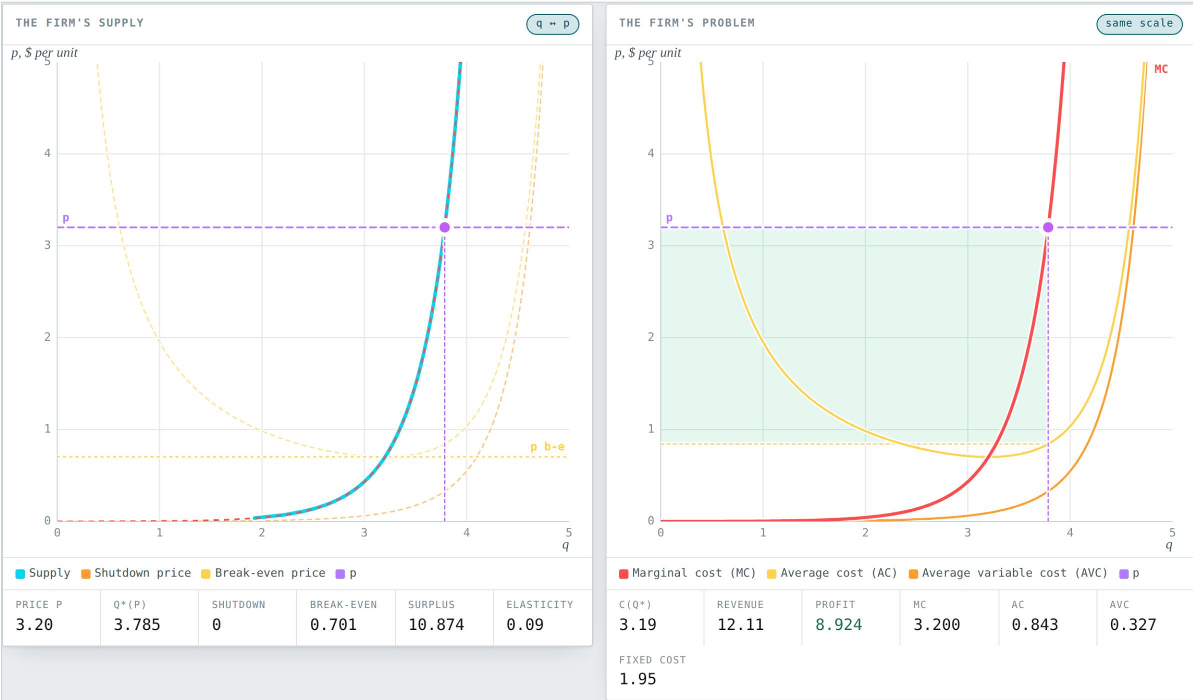


Figure 6. The logistic family at $p = 3.2$. The capacity $\bar{Q} = b\bar{K}^{\gamma}$ is marked by a grey vertical, and marginal cost climbs towards it without ever arriving.

6.2 Firm panel

View	What it draws
Marginals	MC, AC and AVC, in money per unit. That is the same unit as the price, so this panel shares *both* axes with the supply panel and the two figures are comparable pixel for pixel.
Totals	$c(q)$, revenue pq and profit $\pi(q)$, in money. The vertical axis is no longer the supply panel's; the quantity axis still is. The pill in the panel header says which of the two cases you are in.

6.3 The layers

The list changes with the view, because not every layer makes sense in both.

In marginals

Layer	What it draws
Marginal cost	$MC(q)$, in red. It also appears laid over the supply panel.
Average cost	$AC(q)$, in gold. Its minimum is the break-even price.
Average variable cost	$AVC(q)$, in orange. Its minimum is the shutdown price.
Price line	The horizontal at p.
Variable-cost area	The region under MC out to q^*, which is $VC(q^*)$.
Producer surplus area	The region between p and MC out to q^*.
Optimum	The point (q^*, p), its vertical, and the profit rectangle between p and $AC(q^*)$ — green for profit, red for a loss.
Break-even point	Marks the minima of AC and of AVC.
Grid	The background rule.

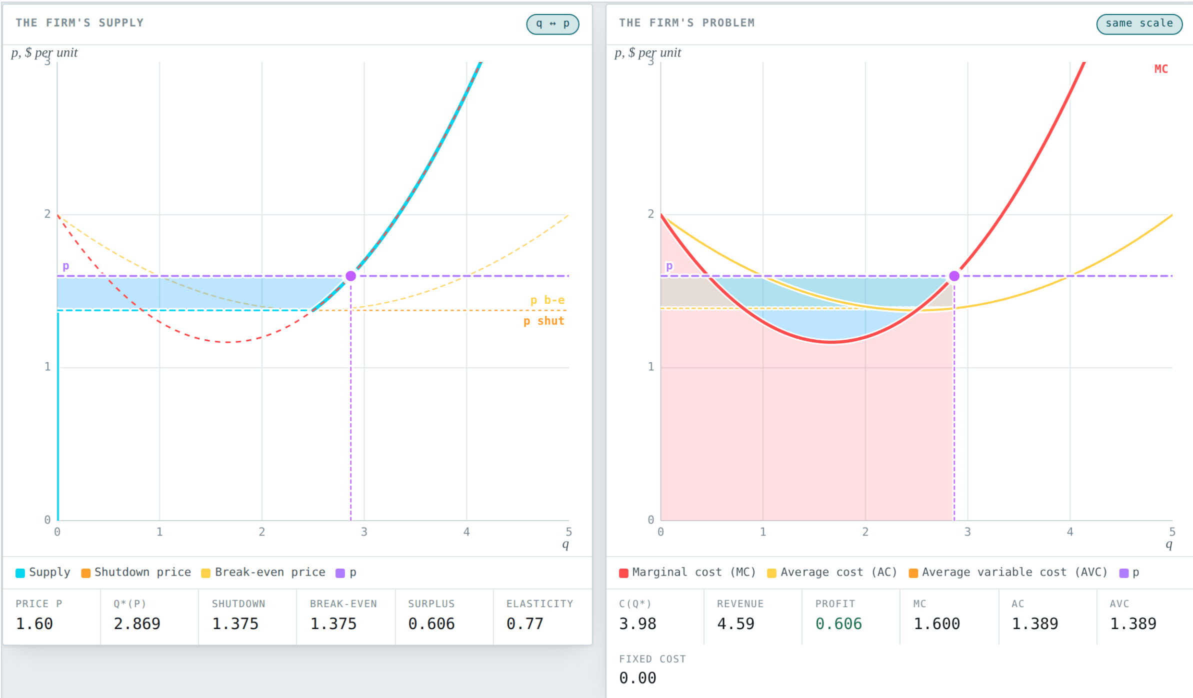


Figure 7. Both areas switched on at once, with the cubic at $p = 1.6$. On the left the surplus measured as the integral of q with respect to price; on the right the same number as the integral of $p - MC$ with respect to quantity.

In totals

Layer	What it draws
Total cost	$c(q)$, with a dotted horizontal at the fixed cost.
Total revenue	The line pq.
Profit	$\pi(q) = pq - c(q)$. Its maximum is at q^*.
Variable cost	$c(q) - F$, the same curve dropped.
Tangent of slope p	The line tangent to $c(q)$ at q^*. It is parallel to the revenue line: there is the first-order condition, drawn.
Optimum	The vertical segment between $c(q^*)$ and pq^*, whose length *is* the profit.
Break-even point	The quantities where $\pi(q) = 0$.

7 How the optimum is found

Worth knowing, because the applet does not do what the GeoGebra original did and the difference shows in the interesting cases.

It does not solve $MC = p$**.** That equation has two roots on a cubic cost curve, and one of them is a minimum of profit. With any non-convex cost function the global optimum can additionally sit at a corner, where no first-order condition finds it.

What it does instead is maximise profit directly:

1. Bound the range from above, by finding the point past which marginal cost has overtaken the price and is still rising.
2. Scan that range with 700 points and keep the best.

Figure 8. The totals view with the cubic at $p = 1.6$. The dotted tangent touches the cost curve exactly where its slope equals the price, and the vertical segment at q^* measures the profit.

3. Refine that neighbourhood by golden section.
4. Compare the result against shutting down, which gives $\pi(0) = -F$. Shutting down is a candidate in its own right, not a limiting case.

The supply curve is traced the same way: a full maximisation at each of 140 prices. It is the only way the jump comes out. Inverting $MC = p$ would never produce it, because below the shutdown price the answer is zero and at the shutdown price it leaps.

> **Note**
>
> The minimum of average variable cost is searched over a geometric ladder, not a linear one: what decides whether the firm ever shuts down is the shape of the curve near zero, and a uniform mesh over five orders of magnitude resolves nothing there. If the minimum lands on an end of the ladder then there is no minimum — the curve is still falling — and the applet reports that there is no shutdown price instead of quoting the smallest value it happened to sample.

8 The verdict

The band at the top names the case and explains it. There are six.

Verdict	When
Produces at a profit	$p > p_{be}$. The price beats minimum average cost.
Produces at a loss	$\underline{p} < p < p_{be}$. And is right to: closing would cost the whole fixed cost.
Zero profit	$p = p_{be}$, the bottom of average cost.
Shuts down	$p < \underline{p}$. The optimum is $q^* = 0$ and the loss is exactly F.
Against the capacity	The optimum is practically at $\bar{Q}$. Only the logistic family reaches this.
No optimum	Marginal cost never overtakes the price over any range worth drawing.

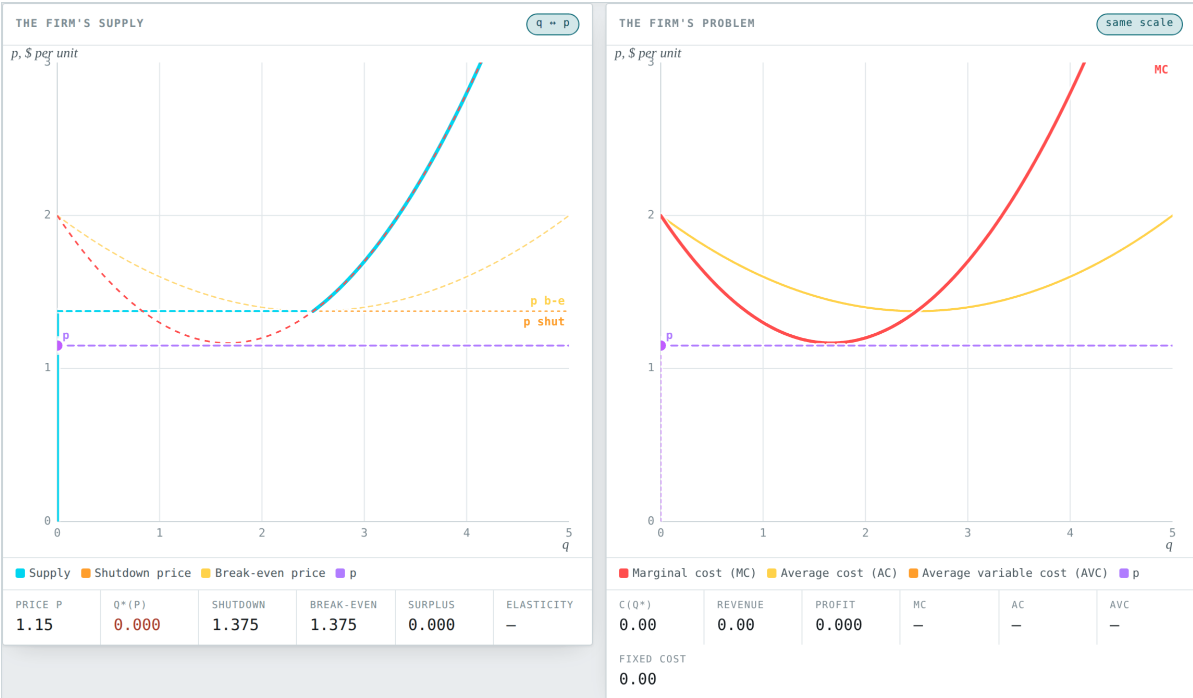


Figure 9. Shutdown, with the cubic at $p = 1.15$. The supply curve lies flat against the price axis, q^* is shown in red, and the verdict explains why producing would be worse than not producing.

9 Classroom activities

Each one starts from the opening settings unless it says otherwise. If you get lost, reload the page.

Activity 1 · The two curves are one

Settings: Cost function **Quadratic / power**, opening values, $p = 2$. Switch **Marginal cost** on if it is off.

What you should see: The cyan curve in the left panel and the red one in the right occupy the same place on the screen. Move p: the purple point travels along both at once and always at the same height. Raise a or k and both steepen together.

Ask the class why the left panel has no average cost curve drawn in bold: because average cost plays no part in the decision of *how much* to produce, only in whether the business is worth being in at all.

Activity 2 · Manufacturing a shutdown price

Settings: Choose **Custom**. The default expression is the cubic.

What you should see: AVC is now U-shaped with minimum 1.375, and supply no longer starts at the origin: there is a stretch flat against the price axis and a jump.

Take the price down to 1.30 and then up to 1.40. The quantity goes from 0 to 2.55 in one step. Ask: what quantity does the firm produce when $p = 1.375$ exactly? The honest answer is that it does not care, and that is why supply there is not a function but a correspondence.

Activity 3 · Producing at a loss

Settings: Cubic. This cubic has $F = 0$, so its shutdown and break-even prices coincide at 1.375 and there is no loss band. Give it a fixed cost: type `1+0.1q^3-0.5q^2+2q` into the box. Set $p = 1.6$.

What you should see: The shutdown price is still 1.375 — fixed cost does not touch average variable cost — but break-even rises to 1.733. There is now a band of prices between them, and 1.6 is inside it.

The **Profit** readout shows −0.394 in red, and still the optimum is to produce 2.869. Ask why it does not close. Closing would cost the whole fixed cost, −1; producing loses only 0.394. Sweep the price across the whole band $1.375 < p < 1.733$: profit is negative throughout and producing stays best.

Activity 4 · Surplus measured twice

Settings: Cubic, $p = 1.6$. Switch on **Producer surplus area**.

What you should see: Two shaded regions of clearly different shapes appear, one in each panel. The **Surplus** readout gives 0.606, and it is both of theirs.

Check it with the numbers in the right panel: $q^* = 2.869$, revenue $= p\,q^* = 4.59$, $c(q^*) = 3.98$. Since $F = 0$ here, variable cost is the whole cost, and $PS = 4.59 - 3.98 = 0.61$. It matches both areas.

Ask why the two regions are not the same shape if they measure the same thing. Because one integrates $p - MC$ with respect to quantity and the other integrates q^* with respect to price; with a jump at the shutdown price they cannot be the same region, but the fundamental theorem of calculus says they have the same area.

Activity 5 · The first-order condition, drawn

Settings: Cubic, $p = 1.6$, firm panel on **Totals**.

What you should see: The dotted tangent to total cost at q^* is parallel to the revenue line. Move the price: the revenue line pivots, and the point of tangency slides along the cost curve to wherever the slope matches again.

The vertical segment at q^* is the profit. Switch on **Break-even point** and the two quantities appear at which that segment has zero length.

Activity 6 · Two roots, one optimum

Settings: Cubic, $p = 1.6$, **Marginals** view.

What you should see: The price line cuts marginal cost *twice*. The purple point is on the right-hand crossing, where MC is rising.

Ask what happens at the other one. It is the solution of $MC = p$ that a student would get by solving the equation without checking the second derivative, and it is a minimum of profit: producing that quantity is the worst the firm can do among those that cover variable cost. Switch to **Totals** to see it as a valley in the profit curve.

Activity 7 · Capacity

Settings: **Logistic log-CD production** family, opening values.

What you should see: The grey vertical marks $\bar{Q} = b\bar{K}^{\gamma}$. Push the price to the top of its slider: the quantity grows by less and less, and the **Elasticity** readout approaches zero.

Now raise $\bar{K}$ and the capacity moves right. That is exactly the long-run exercise the short run forbids: the only way to produce past $\bar{Q}$ is to change plant.

Activity 8 · When there is no optimum

Settings: **Cobb–Douglas production** family. Take β down to 0.05 and s up to 2.

What you should see: The formula in the rail shows $e = 0.526$, and the verdict changes to "No optimum". Every readout in the supply panel turns into a dash.

Marginal cost is falling: each extra unit costs less than the last and sells for the same price. Ask what would happen to such a firm in a competitive market. The answer — that it would grow until it stopped being competitive — is the argument for why diminishing returns are not a technical assumption but the condition for the model to mean anything.

10 Expression syntax

In the cost box the variable is `q`. `x` is accepted too, because that is what GeoGebra called it.

Element	What is allowed
Operators	`+ - * / ^ ( )`
Functions	`sqrt ln log log10 exp abs sin cos tan min max pow`
Constants	`e`, `pi`
Implicit multiplication	`2q`, `3(q+1)`, `2sqrt(q)`
Associativity	`^` binds to the right and tighter than unary minus: `-q^2` is $-(q^2)$

Marginal cost is differentiated symbolically. Where the expression has a kink — `min`, `max`, `abs` — there is no symbolic derivative to be had and it falls back to a central difference, which is what the picture shows anyway.

Watch out

Typing a decreasing cost function, or one with constant marginal cost, lands you in the "No optimum" case. That is not a failure of the applet: that problem has no solution.

11 Changes from the GeoGebra original

Two panels instead of one. The original drew total cost, revenue, profit, marginal cost and average cost on one pair of axes. Money and money per unit share no scale, so the marginal curves were unreadable next to the totals. They are separated here, and the marginal panel is locked to the supply panel's scale.

The supply curve, which was not there. Along with the shutdown price, the break-even price, the jump between them, and the elasticity.

The optimum is a maximisation, not an intersection. The original intersected `Eq: CM(x) = p` with the axis and used the result as the right end of a `Max[Beneficios, 0, x(A)]`. With non-monotone marginal cost that bracket is the wrong one, and when $MC = p$ has no root the whole chain returned `NaN` and the applet went blank.

Shutting down is a candidate. The original always compared against an interior maximum. Here $\pi(0) = -F$ competes with it, which is what makes the shutdown price exist.

An infimum is not a minimum. Average variable cost that merely decreases towards zero has no minimum, so there is no shutdown price. Reporting the smallest sampled value gave a shutdown price of 7.7×10^{-4}.

Sliders bounded away from division by zero. $\beta = 1$ divides by $1 - \beta$; $\bar{K} = 0$ raises zero to a negative power; $w = 0$ and $a = 0$ make cost constant; $k = 1$ makes marginal cost constant. All were reachable in the original.

m **widened** from $[0, 1]$ to $[-1, 2]$, where it actually moves the cost level enough to see.

Bilingual, and with a verdict. Every state of the problem is named and explained, because the shapes on their own do not say which of the six cases a student is looking at.

Dead objects not carried over. `AreaProfi`, `CM2` (a copy of `CM`), `eq1`, `A`, `prueba` and `prueba2` were plumbing for the GeoGebra construction rather than objects with meaning.

12 Troubleshooting

Symptom	What is happening
The panels are empty	The cost expression does not parse. The red message under the box says whether it is a syntax problem or a variable problem.
Everything reads "∞" or a dash	You are in the "No optimum" case. Raise k, or s, or β, or switch family.
The scales stop matching	The firm panel is on **Totals**. The pill in its header says so. Go back to **Marginals**.
The axis moves while I drag the price	Deliberate: axes are rounded to round numbers and step in jumps, so that two settings can be compared by eye instead of renumbering continuously.
The price ignores the mouse	You are dragging inside the totals panel, whose vertical axis is money. Use the supply panel or the slider.
The language or theme was lost	They are kept in the browser. In a private window, or after clearing site data, they go back to their defaults.

13 Publishing it for a class

Any static host will do: GitHub Pages, Netlify, a university web directory, even a shared folder served over HTTP. The only thing to upload is the `index.html` file, inside a folder named however you want the address to read.

The repository ships a GitHub Actions workflow (`.github/workflows/pages.yml`) that publishes every applet at once on each push to the default branch. To switch it on, in the repository:

Settings ⟶ *Pages* ⟶ *Source: GitHub Actions*. That is the one step the workflow cannot do for itself, because creating a Pages site needs administration rights that a `GITHUB_TOKEN` is never granted.

> **Note**
>
> To hand the applet to students without reliable internet, send them the `index.html` file directly. It weighs less than a photograph and opens with a double-click.

14 Licence and provenance

The application code is MIT. Use it, change it and pass it on, in class or out of it, with attribution.

This applet is a standalone rewrite of an original GeoGebra applet. The rewrite is not a literal translation: where the original had a construction error, a division by zero reachable with its own sliders, or a dead object inherited from whatever applet it was copied from, this version fixes it and says so. The section *Changes from the GeoGebra original* lists those changes one by one.

Everything is hand-written and dependency-free: the expression parser, the symbolic differentiation, the level-curve tracing and the drawing. In particular there is no `eval`, so what a student types into a text box cannot become code, and a strict content security policy does not break the page.